

Part 2 — Supervised ML Results

Source:

/home/antony/masai/capstone/spam_detection/run/part2_20260716_211304_418317/cleaned_data.csv |
Run dir: /home/antony/masai/capstone/spam_detection/run/part2_20260716_211304_418317

Feature Matrix

12 features used for both models (shared, leak-free): avg_word_len, capital_ratio, digit_count, punct_count, exclaim_count, digit_level_enc, weekday_Monday, weekday_Saturday, weekday_Sunday, weekday_Thursday, weekday_Tuesday, weekday_Wednesday

char_count was deliberately excluded ($r=0.996$ with the regression target word_count — near-definitional leakage). Categorical features weekday (one-hot, drop-first) and digit_level (ordinal label-encoded Low<Medium<High) were derived specifically to demonstrate both encoding strategies without leaking either target.

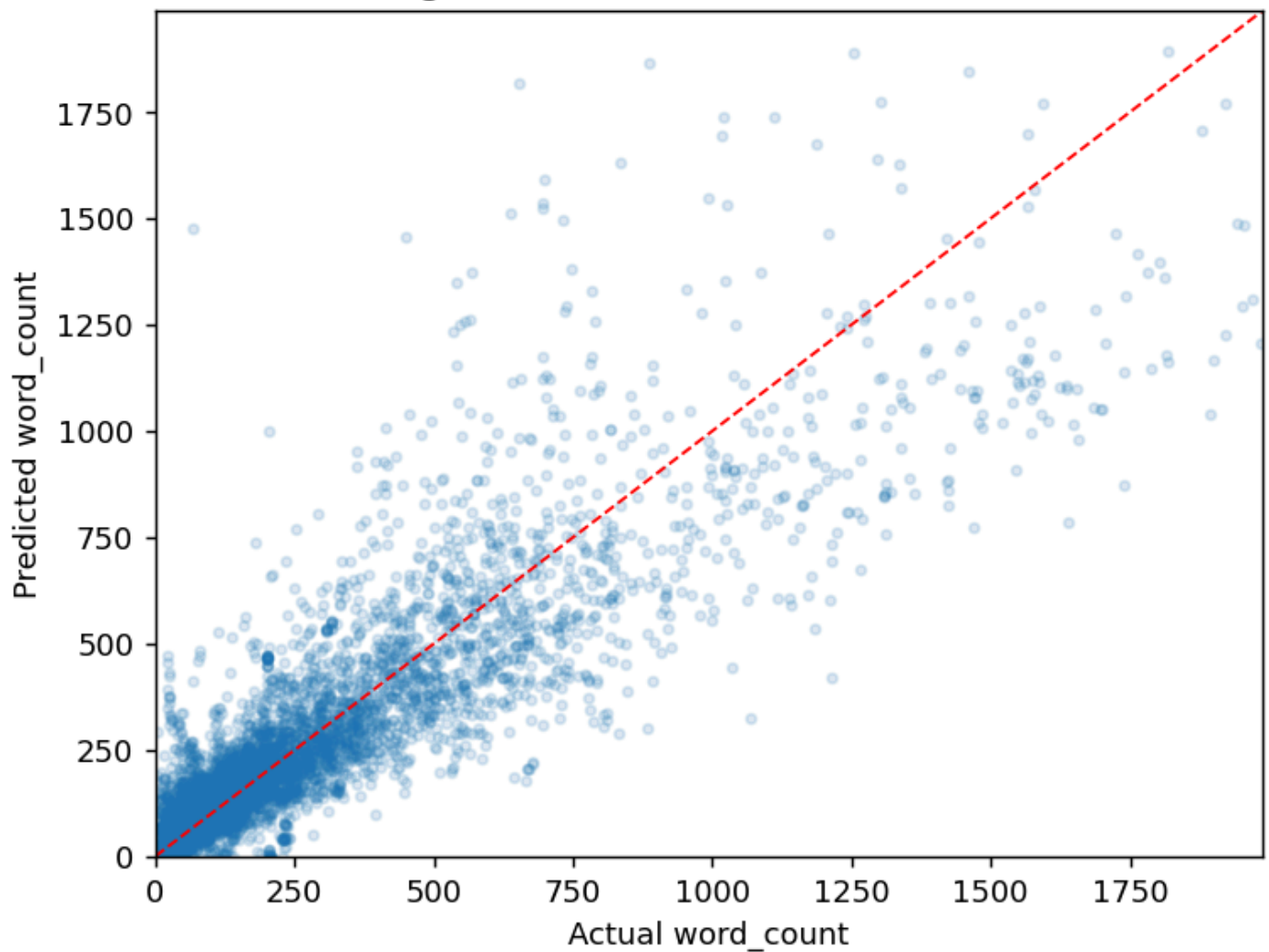
Regression — Predicting word_count

Model	MSE	R ²
Linear Regression	102513.36	0.8584
Ridge (alpha=1.0)	102524.33	0.8584

Top 3 coefficients by |magnitude| (Linear Regression, scaled features)

Feature	OLS coef	Ridge coef
punct_count	612.053	612.008
digit_count	191.111	191.136
avg_word_len	58.437	58.436

Linear Regression: Actual vs Predicted ($R^2=0.858$)



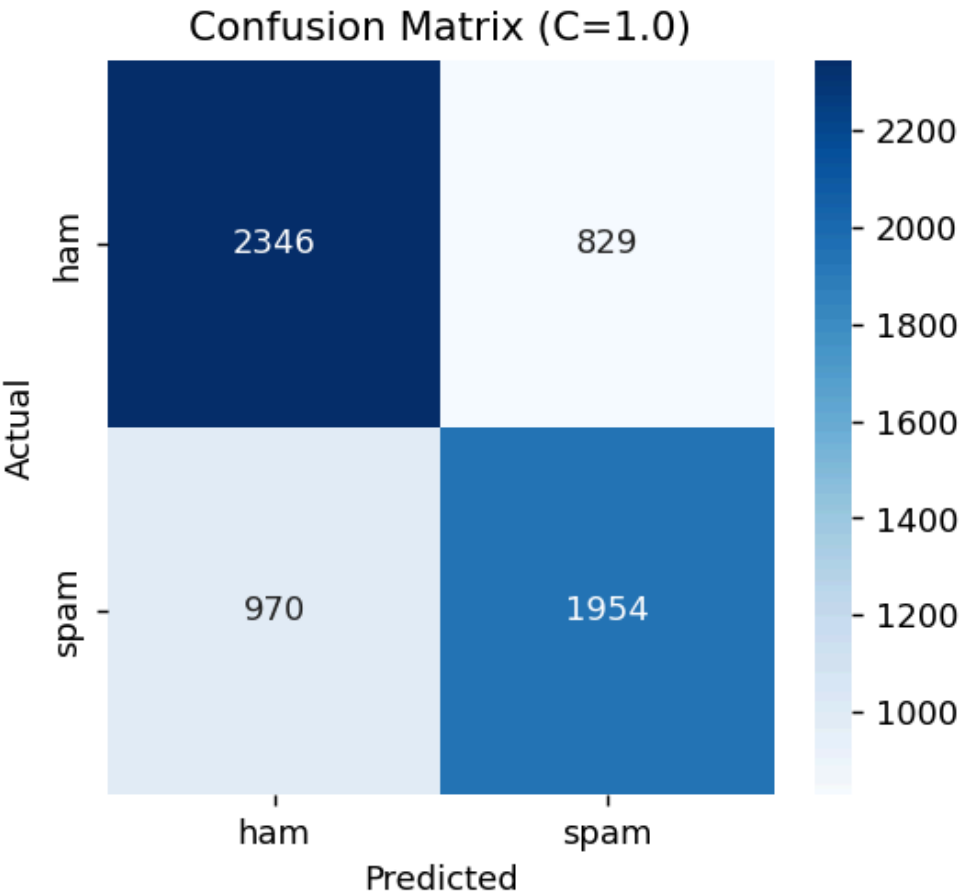
Classification — Spam/Ham

Class balance check (training set)

Class	Count	%
ham	12735	52.2%
spam	11660	47.8%

Minority class = 47.8% (threshold: 35%). Above threshold — no resampling required.
`class_weight='balanced'` applied anyway as a zero-cost weighting safeguard (no synthetic rows, so before/after class counts are identical).

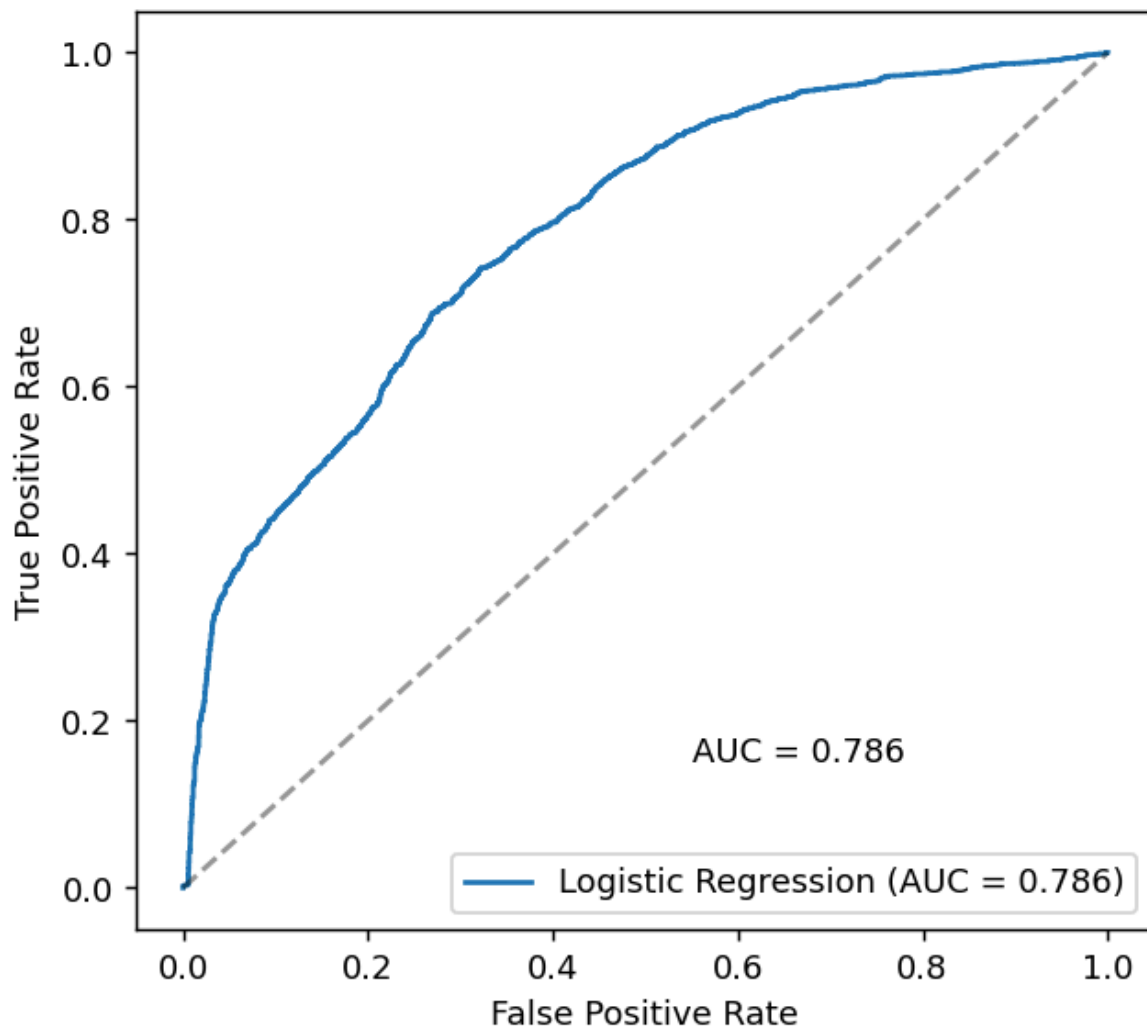
Confusion matrix & classification report ($C=1.0$, threshold=0.5)



Class	Precision	Recall	F1	Support
ham	0.707	0.739	0.723	3175
spam	0.702	0.668	0.685	2924

Accuracy: 0.7050 | AUC: 0.7857

ROC Curve - Spam Classification (Part 2)



Threshold sensitivity

Threshold	Precision	Recall	F1
0.3	0.6011	0.9097	0.7239
0.4	0.6514	0.7900	0.7141
0.5	0.7021	0.6683	0.6848
0.6	0.7505	0.5154	0.6111
0.7	0.8696	0.3694	0.5185

F1-maximizing threshold: **0.3**

Regularization experiment (C=1.0 vs C=0.01)

Model	Precision	Recall	AUC
C=1.0	0.7021	0.6683	0.7857
C=0.01	0.7021	0.6672	0.7837

Bootstrap AUC difference (C=1.0 minus C=0.01, n=500)

Mean difference: **0.0020** 95% CI: [0.0015, 0.0026] Excludes zero: **True**

[← Back to upload](#)